EX. No: 01	TO DEVELOP AN IOT PROGRAM TO TURN ON/OFF LED LIGHT (3.3V)
DATE: 26.12.2025	

AIM:

To create an IOT program to turn on/off LED Light(5V).

ALGORITHM:

Requirements:

1. **Arduino Board** (e.g., Arduino Uno, Nano, etc.)
2. **Push Button**
3. **Resistor** (optional, depending on button wiring)
4. **Jumper Wires**
5. **Breadboard** (optional, for easy connections)
6. **External Power Supply** (for relay if required, depending on your relay's rating)

Connection:

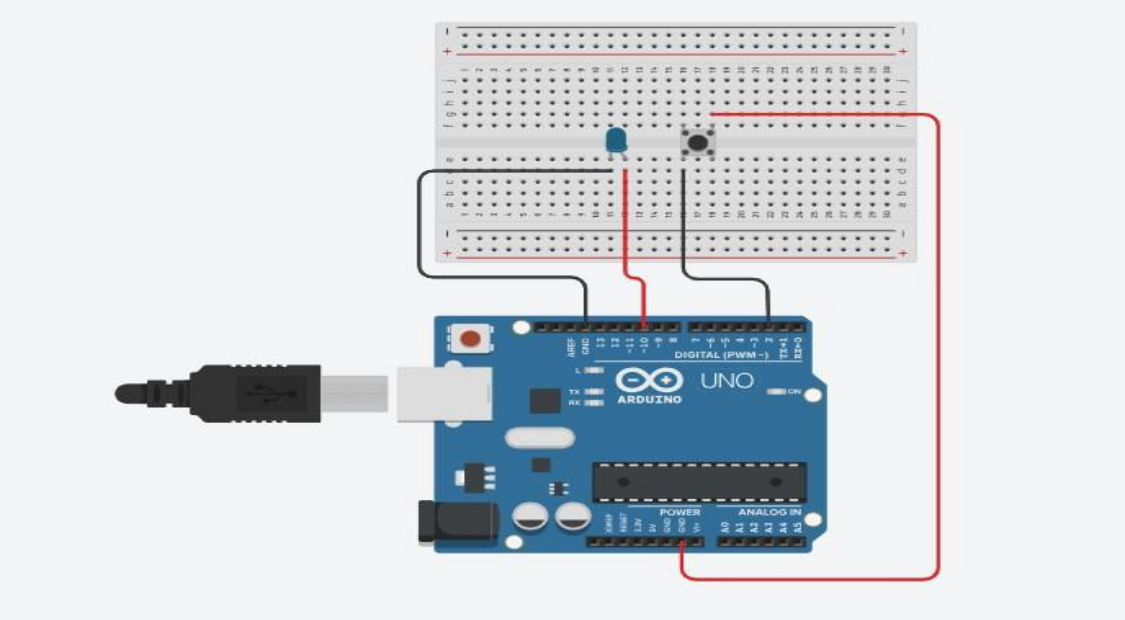
1. Led light (+ive) → pin 10(Arduino)
2. Led light (-ive) → pin GND(Arduino)
3. Pushbutton one side pin → 2(Arduino)
4. Pushbutton another Side Pin → GND(Arduino)

PROGRAM:

```
int pbuttonPin = 2;
int relayPin = 10;
int val = 0;
int prevState = 0;  // To store the previous state of the button
int pushed = 0;

void setup() {
  pinMode(pbuttonPin, INPUT_PULLUP);  // Internal pull-up resistor
  pinMode(relayPin, OUTPUT);          // Set relay pin as output
  digitalWrite(relayPin, HIGH);       // Turn relay off initially
}

void loop() {
  val = digitalRead(pbuttonPin);  // Read the button state

  // Detect a button press event (transition from LOW to HIGH)
  if (val == LOW && prevState == HIGH) {
    pushed = 1 - pushed;          // Toggle the pushed variable
    delay(100);                   // Debouncing delay
  }

  prevState = val;                // Update the previous state

  // Control the relay based on the pushed state
  if (pushed == 1) {
    digitalWrite(relayPin, LOW);  // Turn the relay on
  } else {
    digitalWrite(relayPin, HIGH); // Turn the relay off
  }
}
```

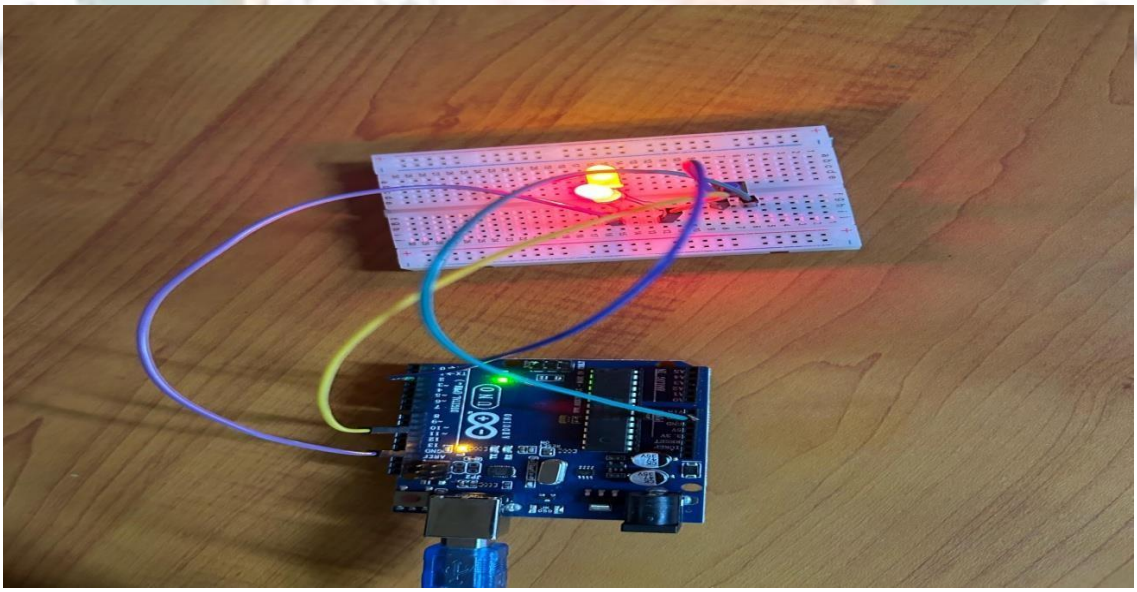
DIAGRAM:

Before didn't click Push Button



OUTPUT:

After Click and Hold Push Button



RESULT:

Thus the above program is verified and executed successfully.

EX. No: 02	TO DEVELOP AN IOT PROGRAM USING IR SENSOR (DETECTING PARKING AVAILABILITY)
DATE: 13.01.2026	

AIM:

To develop an IOT program using IR sensor (Detecting Parking Availability).

ALGORITHM:

Requirements:

1. Arduino UNO Dip,
2. IR sensor module
3. Servo motor,
4. Jumper wires (M-F 10 , M-M 5)
5. LCD Display
6. Beard Board.

Connection:

Arduino to Bread board

- 5v to Positive Rail
- GND to Negative

LCD to Arduino and Bread Board

- GND -Negative Rail
- VCC – Postive Rail
- SDA – A4
- SCL – A5

Arduino to IR sensor 1 and Beard Board

- GND – Negative Rail
- VCC – Positive Rail
- Out – D2

Arduino to IR sensor 2 and Beard Board

- GND – Negative Rail
- VCC – Positive Rail
- Out – D3

Servo motor to Arduino and BreadBoard

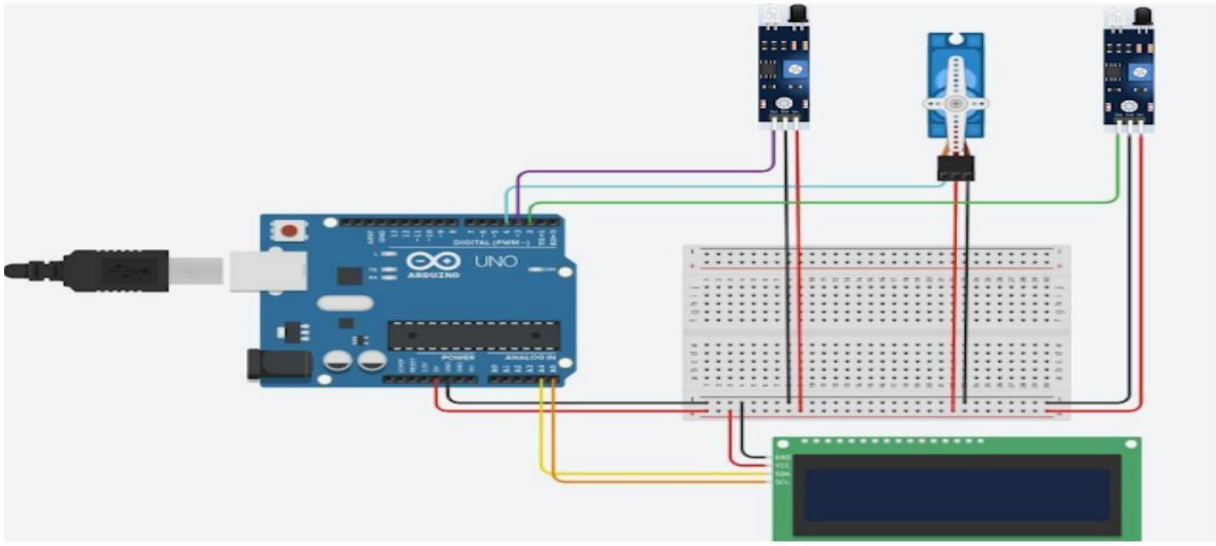
- GND–Negative Rail
- VCC – Positive Rail
- Out – D4

PROGRAM:

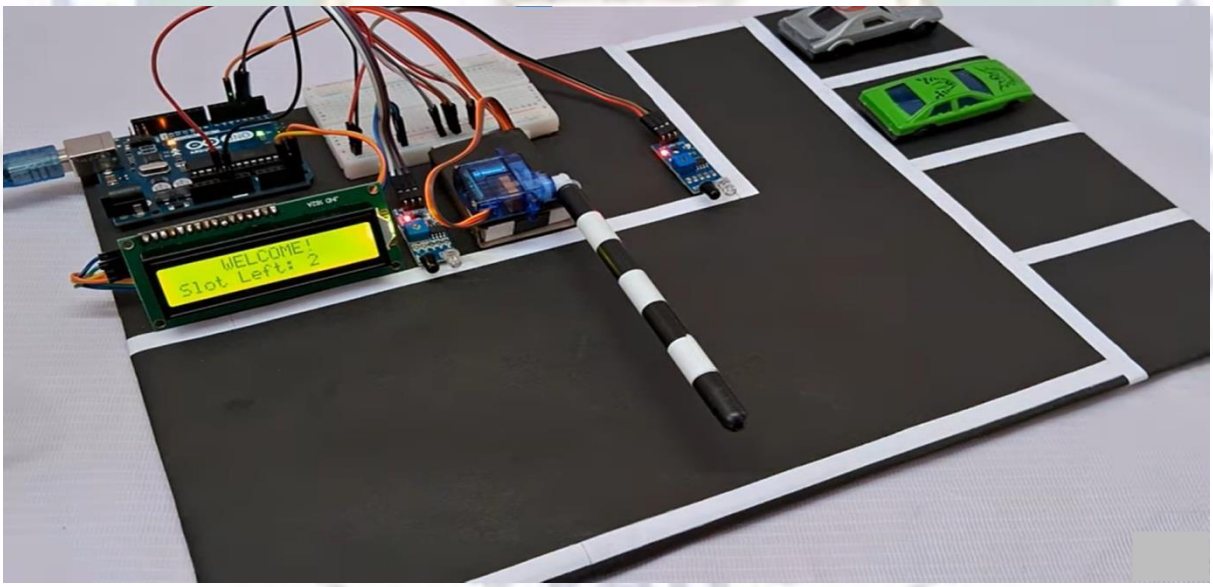
```
#include <Wire.h>

#include <LiquidCrystal_I2C.h>

LiquidCrystal_I2C lcd(0x27,16,2);

#include <Servo.h>

Servo myservo;

int IR1 = 2;

int IR2 = 3;

int Slot = 4;          //Total number of parking Slots

int flag1 = 0;

int flag2 = 0;

void setup() {

  Serial.begin(9600);

    lcd.init(); //initialize the lcd

    lcd.backlight(); //open the backlight

pinMode(IR1, INPUT);

pinMode(IR2, INPUT);

myservo.attach(4);

myservo.write(100);

lcd.setCursor (0,0);

lcd.print("     ARDUINO    ");

lcd.setCursor (0,1);

lcd.print(" PARKING SYSTEM ");

delay (2000);

lcd.clear();

}

void loop(){
```

```
if(digitalRead (IR1) == LOW && flag1==0){
if(Slot>0){flag1=1;
if(flag2==0){myservo.write(0); Slot = Slot-1;}
}else{
lcd.setCursor (0,0);
lcd.print("    SORRY :(    ");
lcd.setCursor (0,1);
lcd.print("  Parking Full  ");
delay (3000);
lcd.clear();
}
}
if(digitalRead (IR2) == LOW && flag2==0){flag2=1;
if(flag1==0){myservo.write(0); Slot = Slot+1;}
}
if(flag1==1 && flag2==1){ delay (1000);
myservo.write(100); flag1=0, flag2=0;
}
lcd.setCursor (0,0);
lcd.print("         WELCOME!     ");
lcd.setCursor (0,1);
lcd.print("Slot Left: ");
lcd.print(Slot);
}
```

DIAGRAM:



OUTPUT:



RESULT:

Thus the above program is verified and executed successfully

EX. No: 03	TO DEVELOP AN IOT PROGRAM USING HUMIDITY AND TEMPERATURE MONITORING.
DATE: 24.01.2026	

AIM:

To develop an iot program using Humidity and temperature monitoring.

ALGORITHM:

Requirements :

1. Arduino board
2. DHT11 sensor
3. 16x2 LCD with I2C
4. Jumper wires (M to F = 7)
5. **Libraries:**
   - LiquidCrystal_I2C for controlling the LCD
   - DHT11 by Adafruit for interfacing with the DHT sensor

Circuit Connections:

Lcd_i2c to arduino:

1. GND - GND
2. VCC - 5V
3. SDA - A4
4. SCL - A5

DHT11 to Arduino:

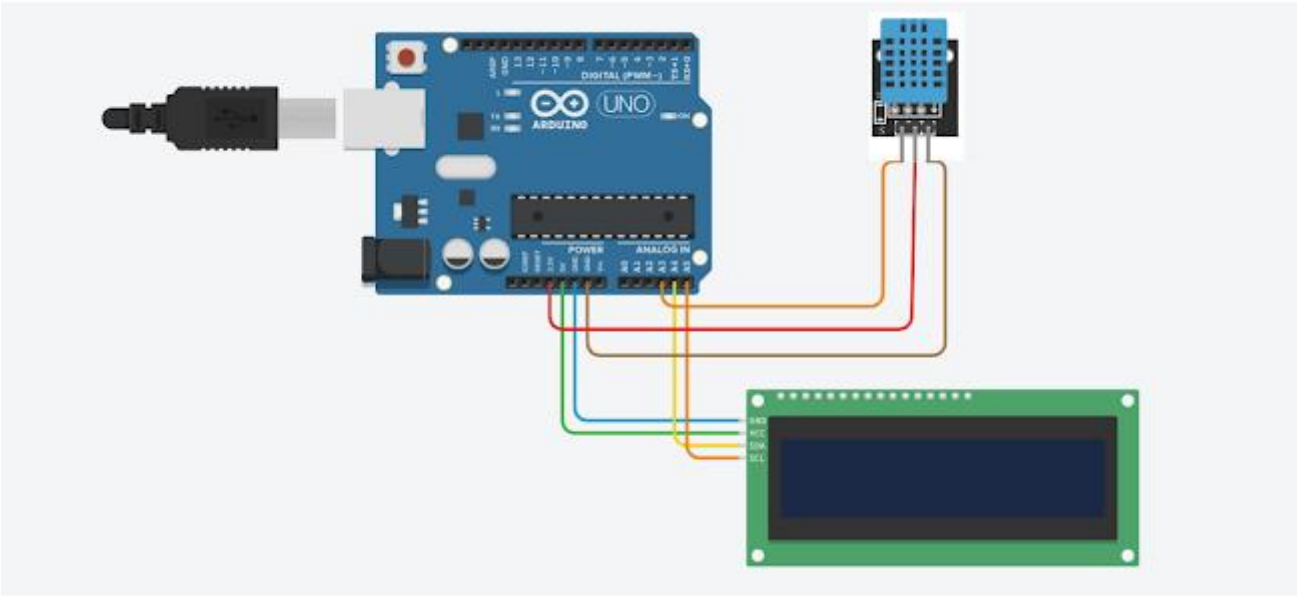
1. VCC - 3.3V
2. DATA - 11
3. GND – GND

PROGRAM:

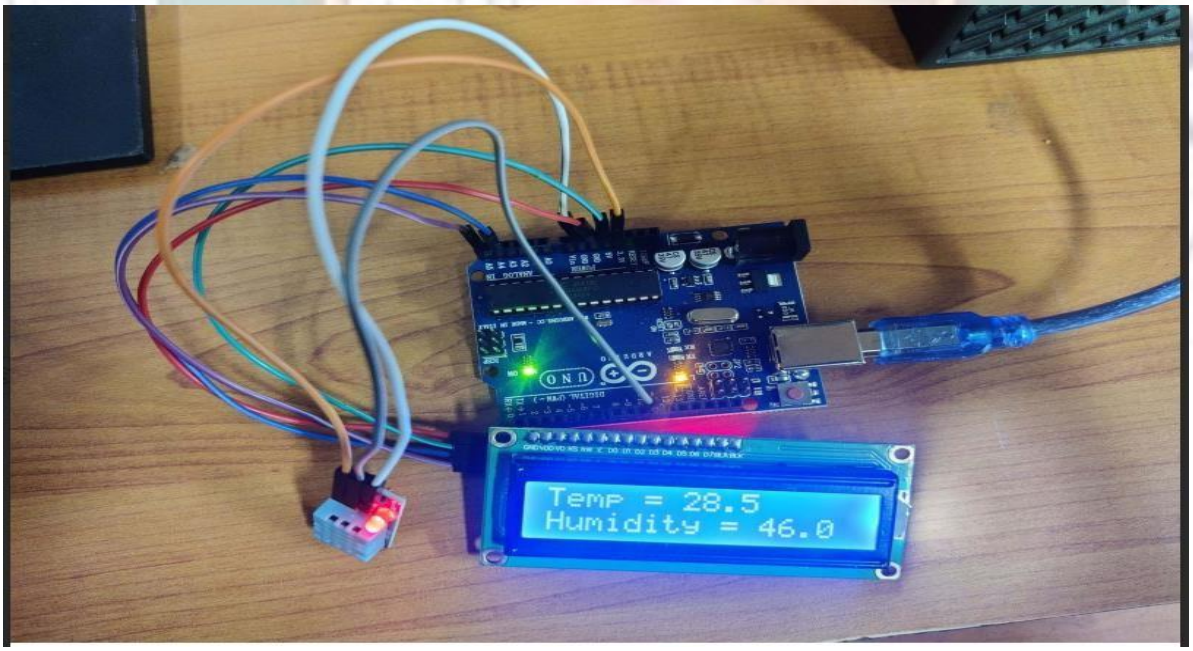
```
//Libraries
#include <DHT.h>
//I2C LCD:
#include <LiquidCrystal_I2C.h>
#include <Wire.h>

LiquidCrystal_I2C lcd(0x27,16,2); // set the LCD address to 0x27 for a 16 chars and 2 line display

//Constants
#define DHTPIN A3     // what pin we're connected to
#define DHTTYPE DHT11   // DHT 11
DHT dht(DHTPIN, DHTTYPE); //// Initialize DHT sensor for normal 16mhz Arduino

//Variables
//int chk;
int h;  //Stores humidity value
int t; //Stores temperature value

void setup()
{
   Serial.begin(9600);
   Serial.println("Temperature and Humidity Sensor Test");
   dht.begin();
   lcd.init(); //initialize the lcd
   lcd.backlight(); //open the backlight
}

void loop()
{
   //Read data and store it to variables h (humidity) and t (temperature)
   // Reading temperature or humidity takes about 250 milliseconds!
   h = dht.readHumidity();
```

```
    t = dht.readTemperature();

    //Print temp and humidity values to serial monitor
    Serial.print("Humidity: ");
    Serial.print(h);
    Serial.print(" %, Temp: ");
    Serial.print(t);
    Serial.println(" ° Celsius");

// set the cursor to (0,0):
// print from 0 to 9:

  lcd.setCursor(0, 0);
    lcd.println(" Simple Circuits");

    lcd.setCursor(0, 1);
    lcd.print(" T:");
    lcd.print(t);
    lcd.print("C");

    lcd.setCursor(11, 1);
    lcd.print("H:");
    lcd.print(h);
    lcd.print("%");

  delay(1000);
```

DIAGRAM:



OUTPUT:



RESULT:

Thus the above program is verified and executed successfully.

EX. No: 04	TO DEVELOP AN IOT WEB SERVER PROGRAM FOR LOCAL HOSTING
DATE: 03.02.2026	

AIM:

To Develop an IOT WEB program for Local Hosting.

ALGORITHM:

Step1: Take the required Components.

- ARDUINO
- WFI Module
- JUMPER WIRE (M –M,M-F)
- RESISTOR
- USB CABLE
- LED light.

Step2: Connect the LED with Resistor in a Pin of 12 And GND.

Step3: Get link the TX in module with Pin of D2, link the RX in module with Pin of 3.

Step4: Join Power line with 3.3Vwith Power module,Connect GND and a power module through a resistor.

Step5: Connect the Arduino to Your System using USB cable.

Step6: Insert and Upload your Arduino Code and Run the Program.

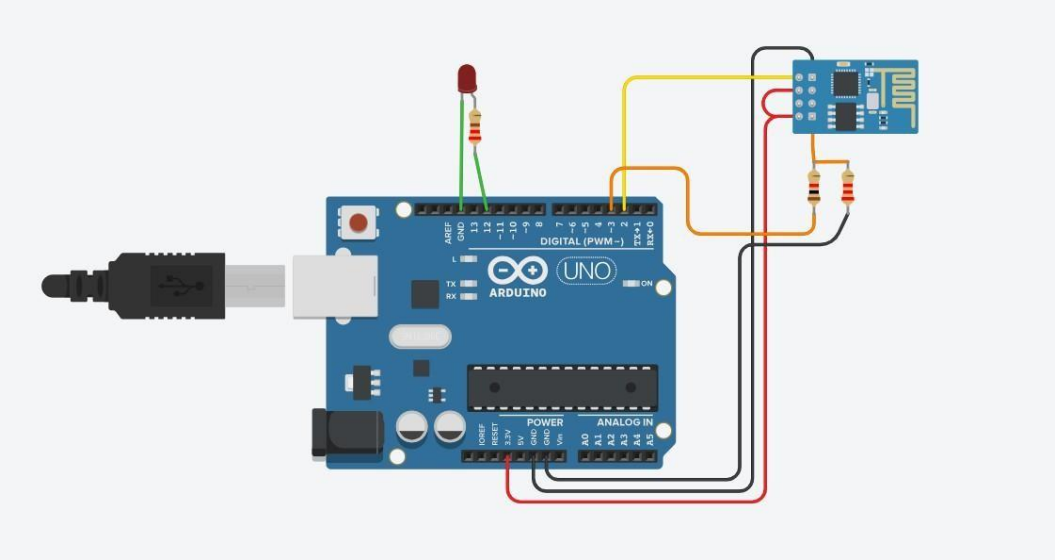
PROGRAM:

```
#include <SoftwareSerial.h>

#define DEBUG true SoftwareSerial esp8266(2,3);

int LED=12; //LED PIN

int itsONled[]={0,0};//para encender el led

void setup()

{

pinMode(LED,OUTPUT); Serial.begin(9600);

esp8266.begin(115200);//depende del modulo esp wifi sendData("AT+RST\r\n",2000,DEBUG);

//sendData("AT+CWMODE=1\r\n",1000,DEBUG);//Configura la salid como access point mode

sendData("AT+CWMODE=1\r\n",1000,DEBUG); //
sendData("AT+CWJAP=\"yourssid\",\"pass\"\r\n", 6000, DEBUG);
sendData("AT+CIFSR\r\n",2000,DEBUG); // para declarar la ip
sendData("AT+CIPMUX=1\r\n",1000,DEBUG);//configure formultiple connections
sendData("AT+CIPSERVER=1,80\r\n",1000,DEBUG); // abre puerto 80 de internet

}

void loop()

{


if(esp8266.available())//chequear el envio del modulo ESP

{

if(esp8266.find("+IPD,"))

{

int connectionId=esp8266.read()-48;

//lectura del string enviado al cliente String msg;

esp8266.find("?"); delay(100);

msg = esp8266.readStringUntil(''); String command1=msg.substring(0);


//CODIGO HTML

Stringwebpage="<html><head><title>Atencion24/7</title>";

webpage += "<meta name=\"viewport\" content=\"width=device-width, initial-
scale=1.0\"><style>.button{background-color:orange;border:none;color:white;padding: 15px
32px;text-align: center;display: inline-block;font-size: 16px;} .centre {text-align: center;}</style>";
```

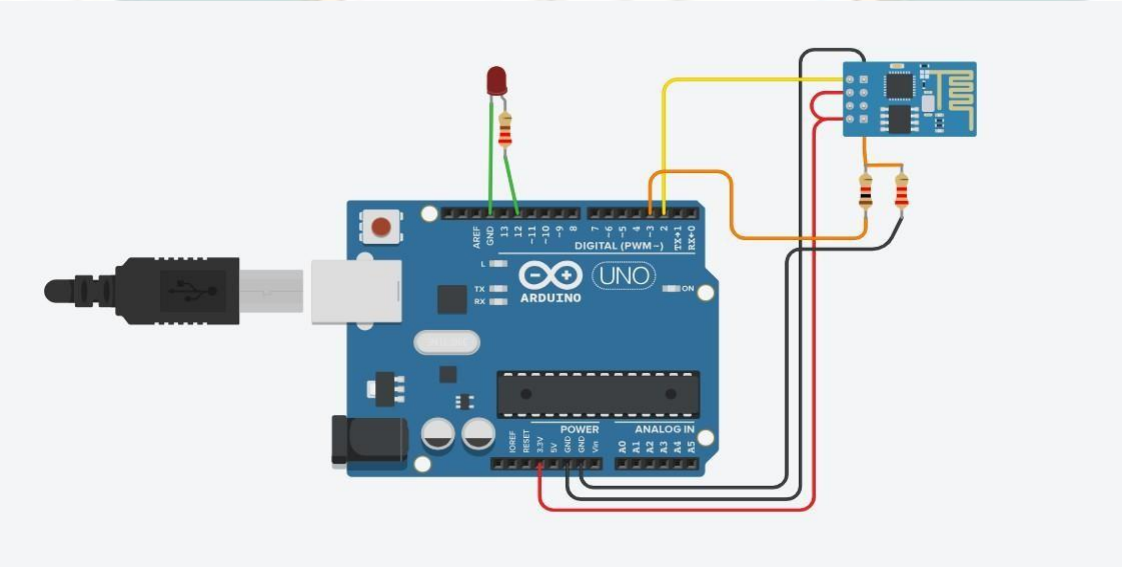
```
webpage+="</head><bodyclass=\"centre\"><h1 class=\"centre\">Bienvenido</h1>";

//COMNADO PARA ENCENDER Y APAGAR LED
if(command1=="T"){ if (itsONled[1] == 1)
{
digitalWrite(LED,LOW); itsONled[1] = 0; webpage+="<p>apagado</p>";
}
else
{
digitalWrite(LED,HIGH); itsONled[1] = 1; webpage+="<p>encendido</p>";
}
}
webpage+="<aclass=\"button\"href=\"?T\">TAP</a></body></html>"; String cipSend =
"AT+CIPSEND=";
cipSend+=connectionId;
cipSend+=",";
cipSend +=webpage.length(); cipSend +="\r\n"; sendData(cipSend,500,DEBUG);
sendData(webpage,500,DEBUG);

//BELOWTHISLINECLOSETHECONNECTION
String closeCommand = "AT+CIPCLOSE="; closeCommand+=connectionId;//append connection id
closeCommand+="\r\n"; sendData(closeCommand,500,DEBUG);
}
}
}
//enviodedatosporelmoduloESP
void sendData(String command,const int timeout,boolean debug)
{
esp8266.print(command);//send the read character to the esp8266 long int time = millis();
while((time+timeout)>millis())
{
while(esp8266.available())
```

```
{
//Para que se vea por el monitor serial Serial.write(esp8266.read());
}
}
}
```

DIAGRAM:



OUTPUT:



RESULT:

Thus the above program is verified and executed successfully.

EX. No: 05	TO DEVELOP AN IOT PROGRAM USING SOIL MOISTURE SENSOR
DATE: 11.02.2026	

AIM:

To develop an IOT program using soil moisture sensor.

ALGORITHM:

Requirements:

1. Arduino Board
2. Soil moisture Sensor
3. Jumper Wire(M to F=3)

Circuit Connections:

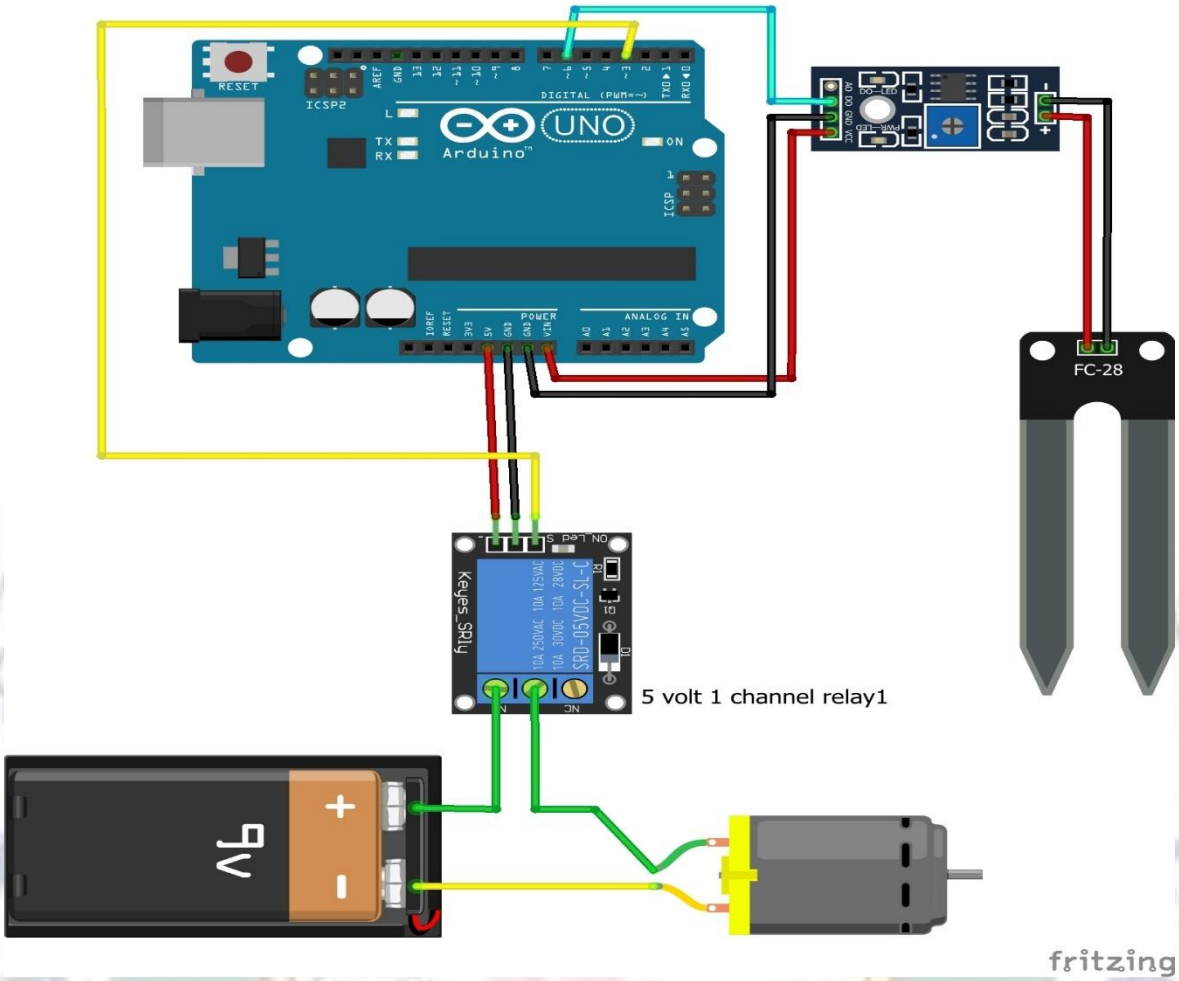
Arduino to soil moisture sensor:

1. GND – GND
2. 5V – VCC
3. A0 – A0

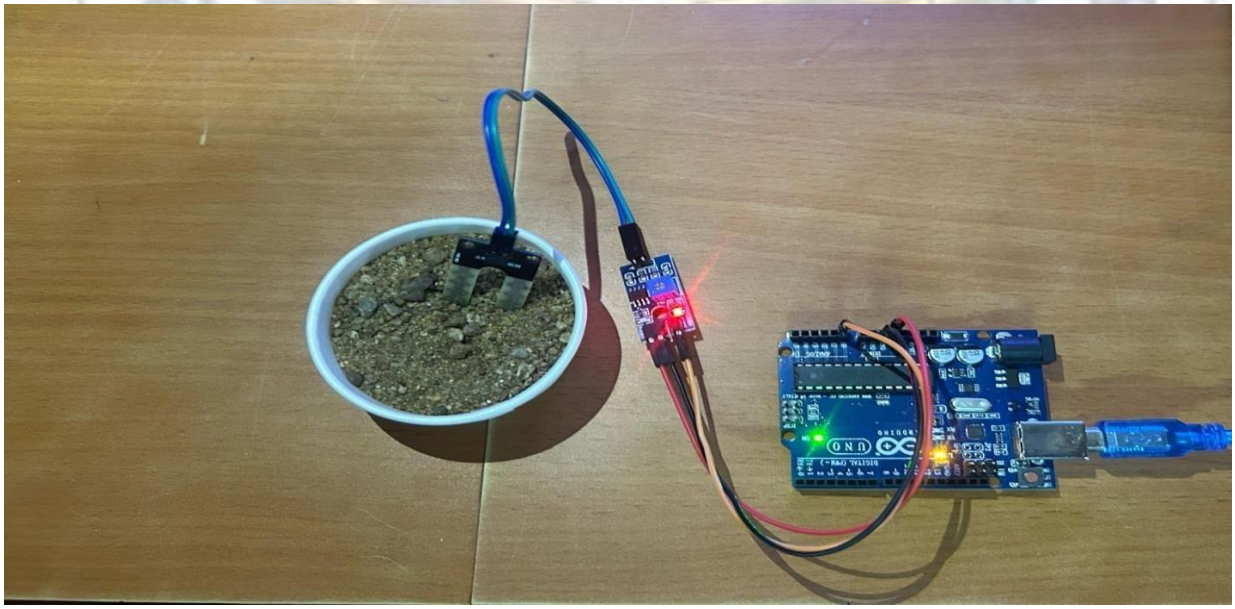
PROGRAM:

```
int water; //random variable
void setup() {
  pinMode(3,OUTPUT); //output pin for relay board, this will sent signal to the relay
  pinMode(6,INPUT); //input pin coming from soil sensor
}
void loop() {
  water = digitalRead(6);  // reading the coming signal from the soil sensor
  if(water == HIGH) // if water level is full then cut the relay
  {
  digitalWrite(3,LOW); // low is to cut the relay
  }
  else
  {
  digitalWrite(3,HIGH); //high to continue proving signal and water supply
  }
  delay(400);
}
```

DIAGRAM:



OUTPUT:



RESULT:

Thus the above program is verified and executed successfully.

EX. No: 06	TO DEVELOP AN IOT PROGRAM USING ULTRASONIC SENSOR (DISTANCE MEASUREMENT)
DATE: 19.02.2026	

AIM:

To Develop an iot program using ultrasonic sensor (distance measurment).

ALGORITHM:

APPARATUS NEEDED

1. Arduino (UNO)
2. HC –SR04 Ultrasonic sensor
3. 16x2 LCD I2C Display
4. Bread Board
5. Jumper wire

HC-SR04 ULTRASONIC SENSOR

1. VCC - Connect to Bread Board
2. GND - GND on Arduino
3. Trig - Pin 9 on Arduino
4. Echo - Pin 10 on Arduino

LCD I2C DISPLAY

1. VCC - Connect to Bread Board
2. GND - GND on Arduino
3. SDA - A4 on Arduino
4. SCL - A5 on Arduino

ARDUINO

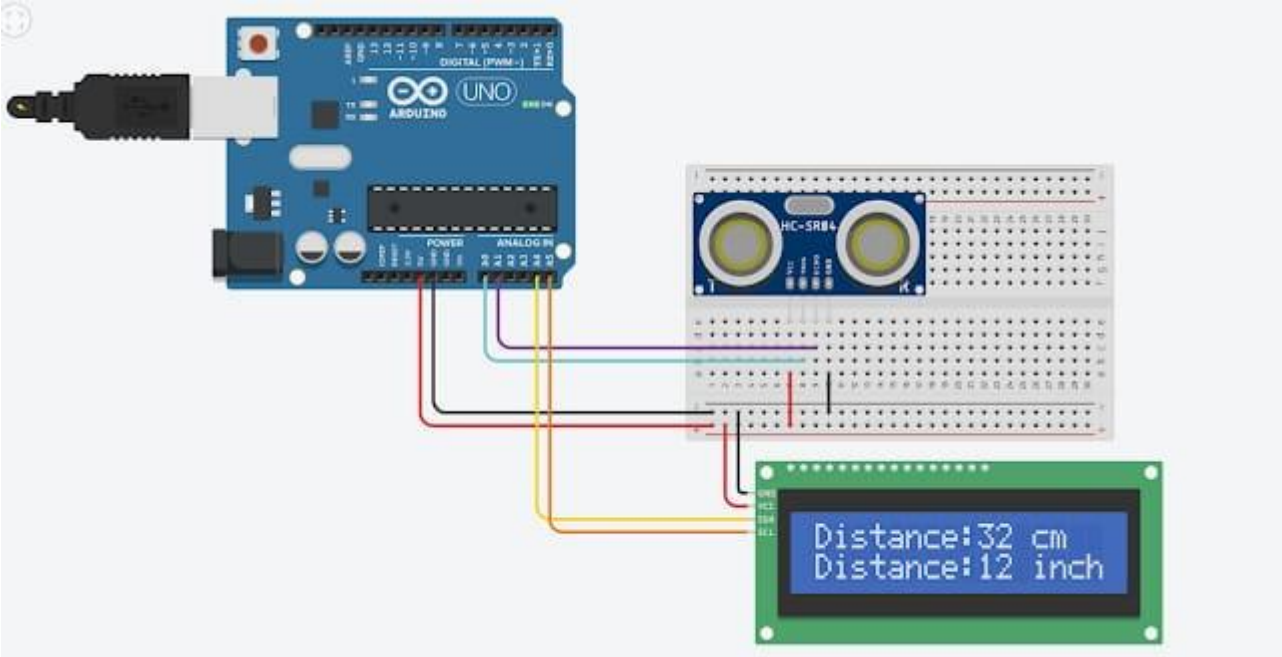
5V - Connect to BreadBoard.

PROGRAM:

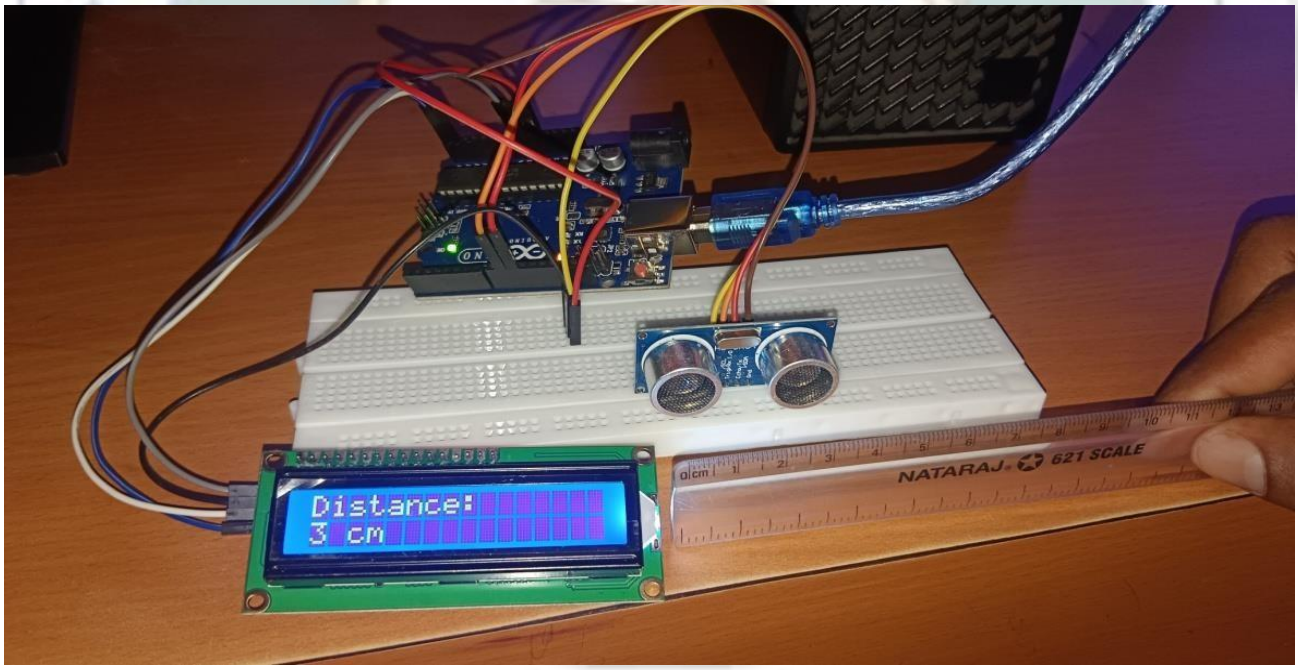
```
#include <Wire.h>

#include <LiquidCrystal_I2C.h>

LiquidCrystal_I2C lcd(0x27, 16, 2);

#define trigPin A0 //Sensor Trig pin connected to Arduino pin A0

#define echoPin A1 //Sensor Echo pin connected to Arduino pin A1

long distanceInch;

void setup()

{

pinMode(trigPin, OUTPUT);

pinMode(echoPin, INPUT);

lcd.init();

lcd.backlight();

lcd.clear();

lcd.setCursor(0,0);

lcd.print("Simple  Circuits");

delay(2000);

lcd.clear();

lcd.setCursor(0,0);  //Set LCD cursor to upper left corner, column 0, row 0

lcd.print("Distance:");//Print Message on First Row

lcd.setCursor(0,1);

lcd.print("Distance:");

}

void loop()

{

long duration, distance;

digitalWrite(trigPin, LOW);

delayMicroseconds(2);
```

```
digitalWrite(trigPin, HIGH);
delayMicroseconds(10);
digitalWrite(trigPin, LOW);
duration = pulseIn(echoPin, HIGH);
distance = (duration/2) / 29.1;
distanceInch = duration*0.0133/2;
lcd.setCursor(9,0);
lcd.print("                    ");
lcd.setCursor(9,0);
lcd.print(          distance); //Print measured distance
lcd.print(" cm");  //Print your units.
lcd.setCursor(9,1);  //
lcd.print("                    "); //Print blanks to clear the row
lcd.setCursor(9,1);
lcd.print(distanceInch);
lcd.print(" inch");  //Print your units.
delay(200); //pause to let things settle
}
```

DIAGRAM:



OUTPUT:



RESULT:

Thus the above program is verified and executed Successfully.

EX. No: 07	TO DEVELOP AN REAL-TIME IOT PROGRAM USING RELAY MODULE (SMART HOME AUTOMATION WITH 230V)
DATE: 26.02.2026	

AIM:

To write a program to develop an real time iot program using relay module (smart home automation with 230v).

ALGORITHM:

COMPONENT REQUIRED:

- Arduino UNO
- Relay
- LDR
- 1k ohm Resistor
- Breadboard
- Jumper Wire(M-F 3 and M-M 5)
- Bulb (230v)
- Bulb Holder

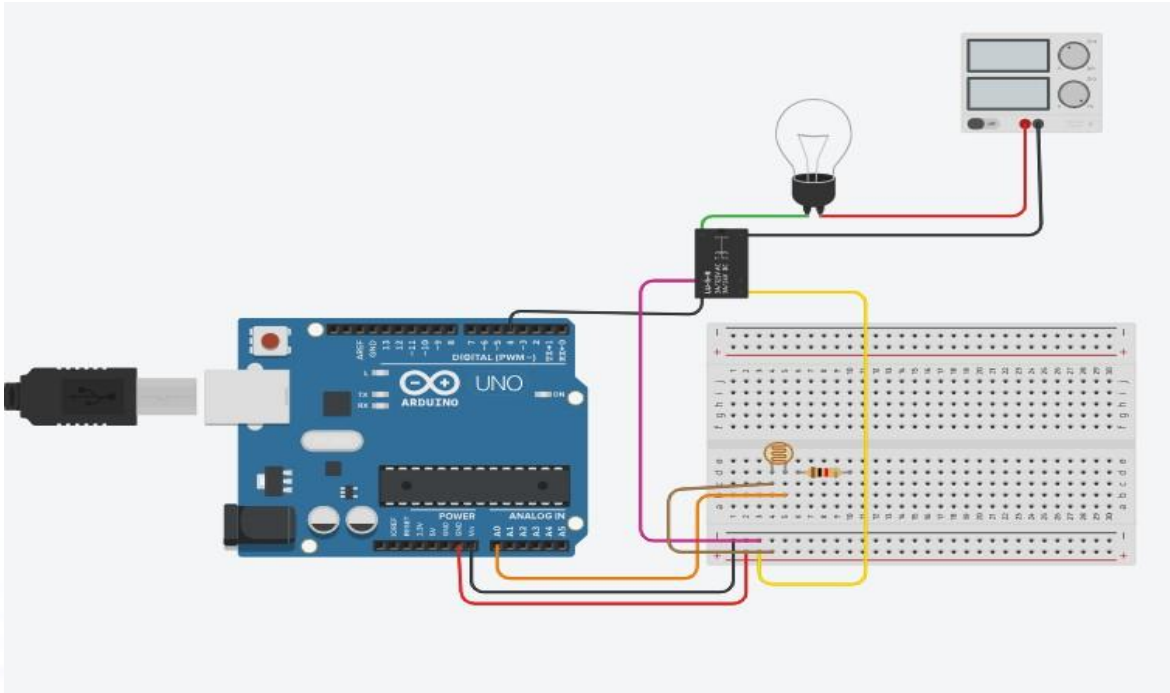
CONNECTIONS:

- Connect LDR with 1k ohm Resistor
- Connect Resistor with –ve of Breadboard
- Connect LDR with +ve of Breadboard
- Connect other LDR side with Pin A0
- Relay vcc with +ve
- Relay GND with –ve
- Breadboard +ve with 5v Arduino
- Breadboard –ve with GND Arduino
- Connect wire AC to Relay COM
- Connect Wire Bulb to Relay NC

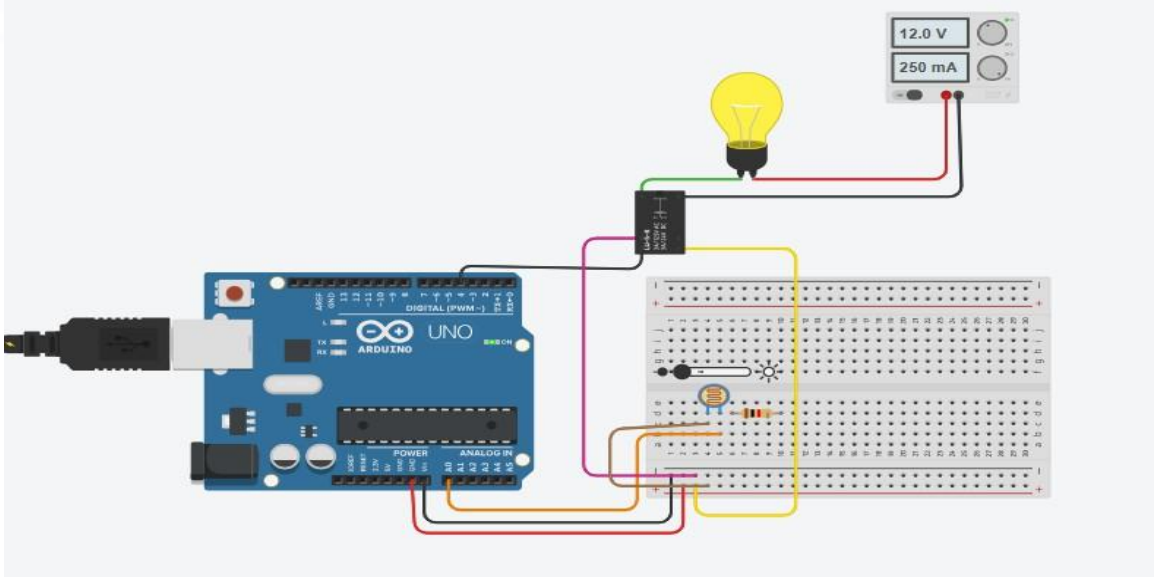
PROGRAM:

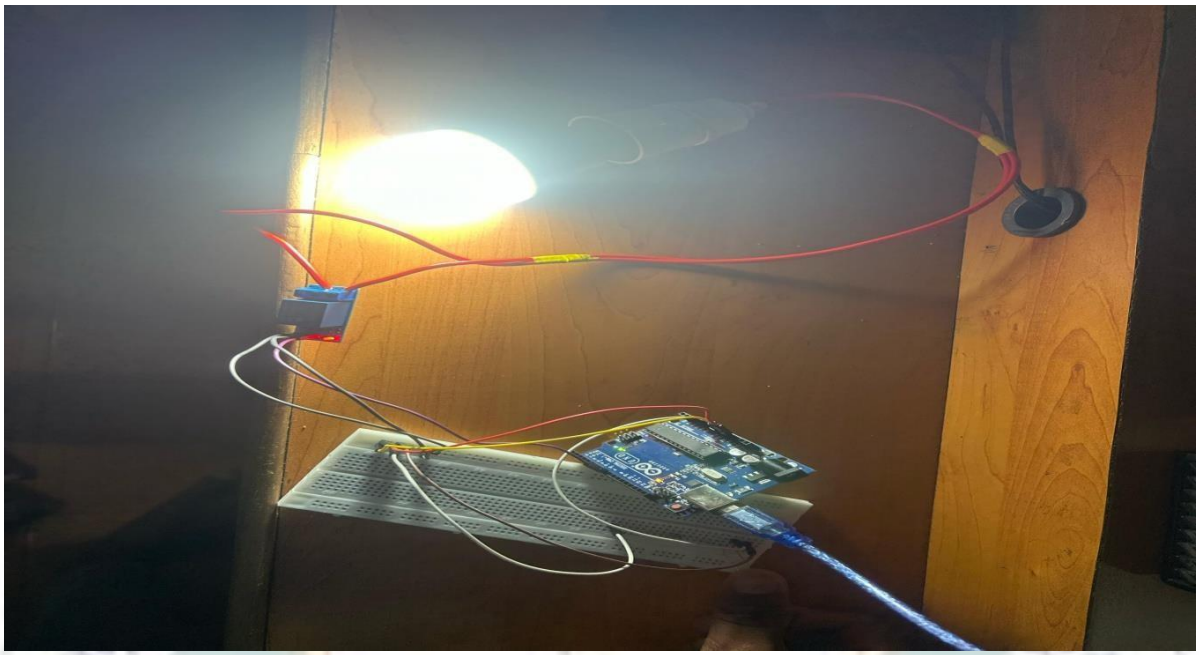
```
int sensorPin = A0;
int relayPin = 8;
void setup()
{
Serial.begin(9600); pinMode(sensorPin, INPUT); pinMode(relayPin, OUTPUT);
}
void loop()
{
int sensorValue = analogRead(sensorPin);
if( sensorValue <= 25 ) // Change the value as per your requirement
{
digitalWrite(relayPin, LOW); Serial.println(sensorValue); delay(100);
}
else
{
digitalWrite(relayPin, HIGH); Serial.println(sensorValue);
}
}
```

DIAGRAM:



OUTPUT:





RESULT:

Thus the above program is verified and executed successfully.

EX. No: 08	TO DEVELOP AN IOT PROGRAM FOR FIRE DETECTION (HOME, INDUSTRY.)
DATE: 26.02.2026	

AIM:

To Develop an IOT program For Fire Detection.

ALGORITHM:

Step 1: APPARATUS NEEDED

- ARDUINO
- FIRE SENSOR MODULE
- JUMPER WIRE (M –M=2 , M-F=3)
- BUZZER
- BREADBOARD
- USB CABLE
- LED

Step 2: Circuit Diagram:

Flame Sensor (analog output):

VCC → 5V (Arduino)

GND → GND (Arduino)

A0 (Analog Output) → A0 (Arduino)

LED:

Long leg (Anode) → Digital Pin 13 (Arduino)
Short leg (Cathode) → GND (Arduino)

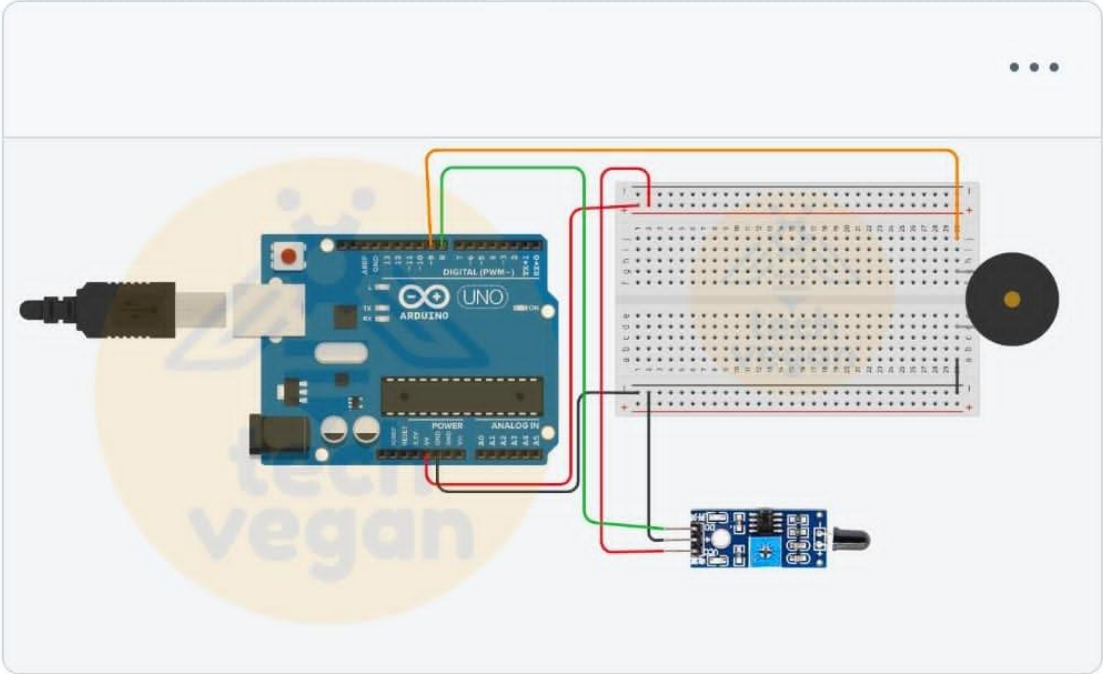
Buzzer:

Positive leg → Digital Pin 12 (Arduino)
Negative leg → GND (Arduino)

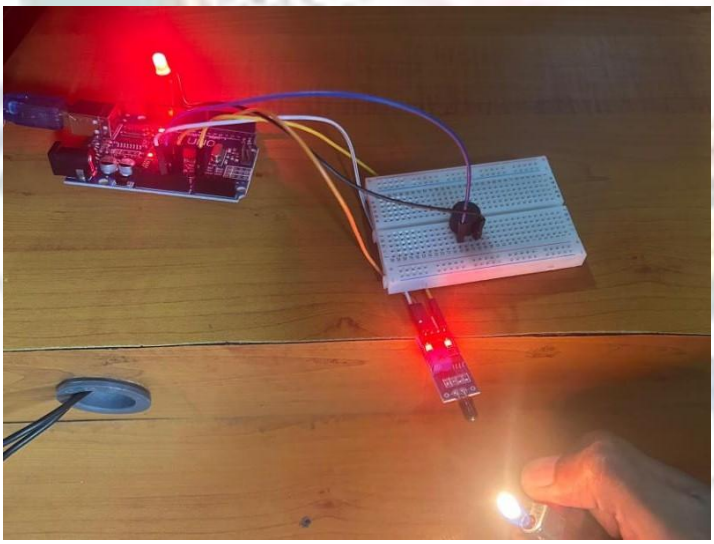
PROGRAM:

```
const int buzzerPin = 9;
const int fireSensorPin = 8;
void setup()
{
  Serial.begin(9600);
  pinMode(buzzerPin, OUTPUT);
  pinMode(fireSensorPin, INPUT);
}
void loop()
{
  int fireValue = digitalRead(fireSensorPin);
  Serial.println(fireValue);
  if(fireValue == 0)
  {
    digitalWrite(buzzerPin, HIGH);
    delay(5000);
  }
  else
  {
    digitalWrite(buzzerPin, LOW);
  }
  delay(500);
}
```

DIAGRAM:



OUTPUT:



```
No Fire
833
No Fire
17
Fire Detected!
794
No Fire
946
No Fire
953
No Fire
---
```

RESULT:

Thus the above program is verified and executed successfully.

EX. No: 09	TO DEVELOP AN IOT PROGRAM FOR GAS LEAKAGE DETECTION (HOME, INDUSTRY.)
DATE: 06.03.2026	

AIM:

To Develop An IOT Program For Gas Sensor Leakage Detection.

ALGORITHM:

Step1:Materials Needed

- Arduino board(e.g.,Arduino Uno)
- MQ2 Gas sensor.
- Resistors(2)[1k ohm]
- Buzzer module
- LED(Red, green)(optional for visual alert)
- Jumper wires
- Breadboard
- Power source (e.g.,battery or USB cable)

Step2:Connection

MQ5 TO Arduino and bread board

- VCC - Positive Rail in Bread Board to 5v in Arduino
- GND – Negative Rail in BreadBoard to GND in Arduino
- OUT – A0(Arduino)

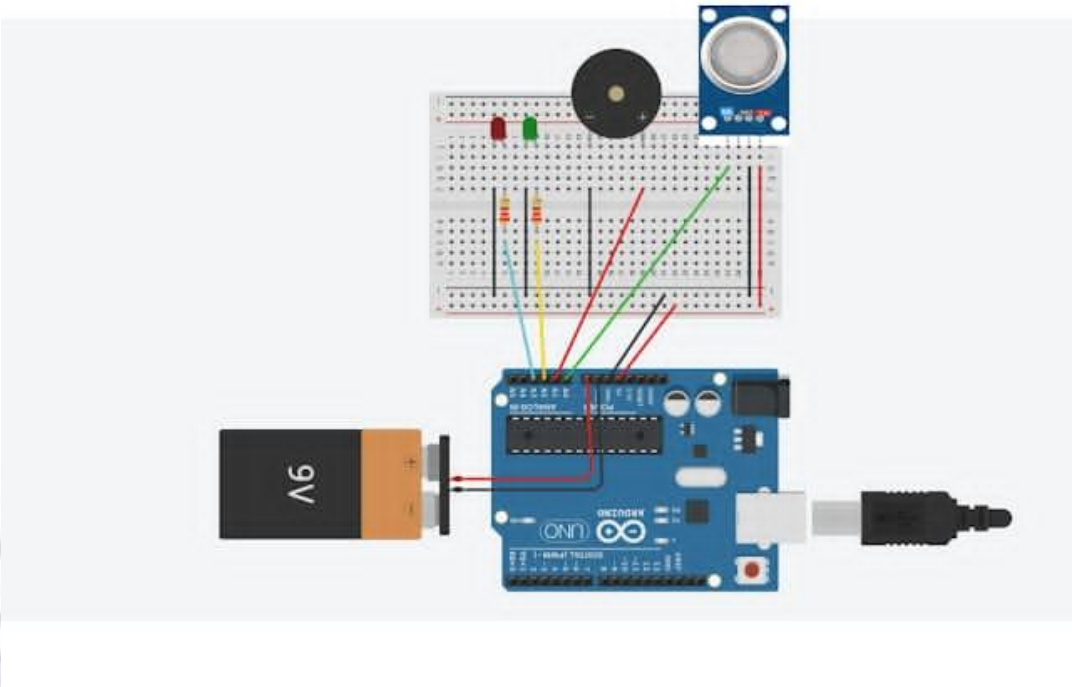
Red LED , Green LED and Buzzer to arduino and BreadBoard

- Red led positive – Pin D2 of Arduino
- Red led Negative – resistor (-ive) Rail to BreadBoard
- Green led positive – Pin D3 of Arduino
- Green led Negative – resistor (-ive) Rail to BreadBoard
- Buzzer Positive – Pin D4 of Arduino
- Buzzer Negative – Negative rail on BreadBoard

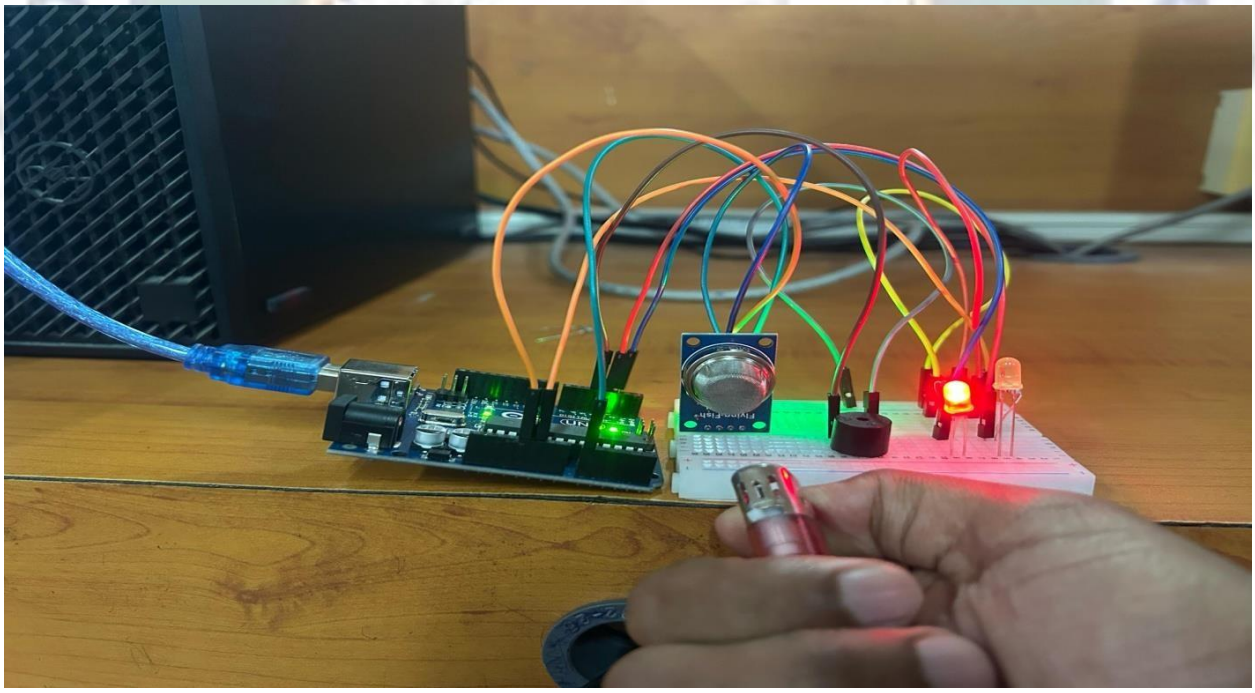
PROGRAM:

```
int Input = A0;
int Buzzer = A1;
int GreenLED = A2;
int RedLED = A3;
int value;
int MAX = 600;
void setup() {
  Serial.begin(9600);
  pinMode(Input ,INPUT);
  pinMode(RedLED ,OUTPUT);
  pinMode(GreenLED ,OUTPUT);
  pinMode(Buzzer ,OUTPUT);
}
void loop() {
  value = analogRead(A0);
  if (value >= MAX) {
    digitalWrite(GreenLED ,LOW);
    digitalWrite(RedLED ,HIGH);
    digitalWrite(Buzzer ,HIGH);
    delay(5000);
    digitalWrite(Buzzer ,LOW);
    delay(6000);
  }
  else {
    digitalWrite(RedLED ,LOW);
    digitalWrite(Buzzer ,LOW);
    digitalWrite(GreenLED ,HIGH);
  }
}
```

DIAGRAM:



OUTPUT:



RESULT:

Thus the above program is verified and executed successfully.

EX. No: 10	TO DEVELOP AN IOT PROGRAM USING HEARTBEAT SENSOR
DATE: 16.03.2026	

AIM:

To Develop An Iot Program Using Heartbeat Sensor.

ALGORITHM:

Requirement:

1. Arduino Uno Board
2. Jumper Wire(M-F 7)
3. 16x2 LCD Display
4. Heartbeat Sensor Module with Probe(finger based)

Connection:

LCD display to Arduino

- GND – GND
- VCC – ICSP 1
- SDA – A5
- SCL – A4

Heart beat sensor to Arduino

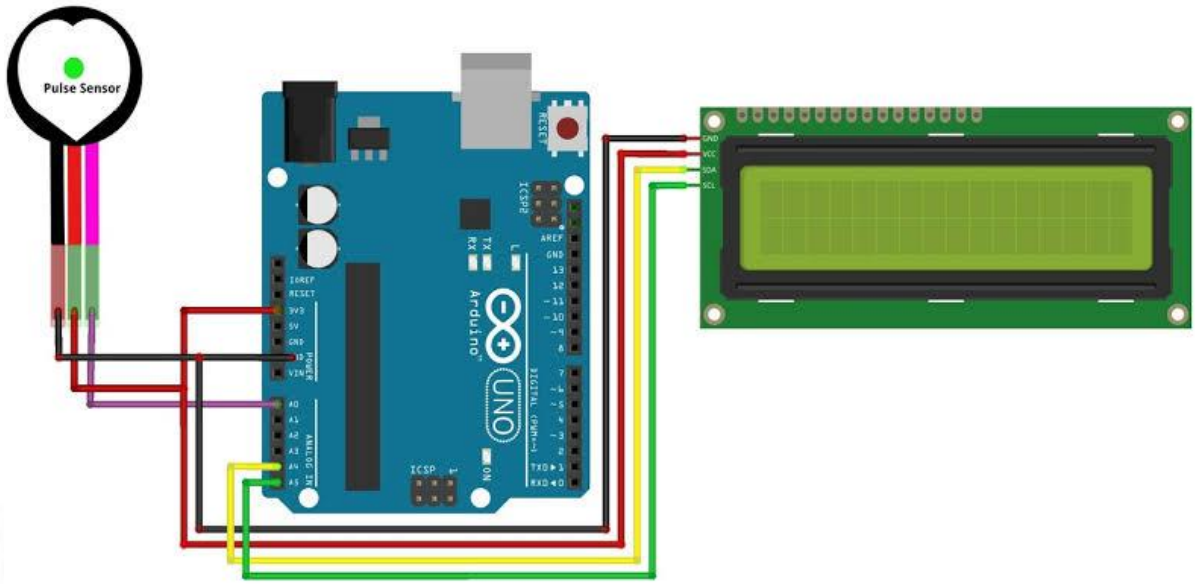
-(ive) – GND
+(ive) – 5v
S – A0

PROGRAM:

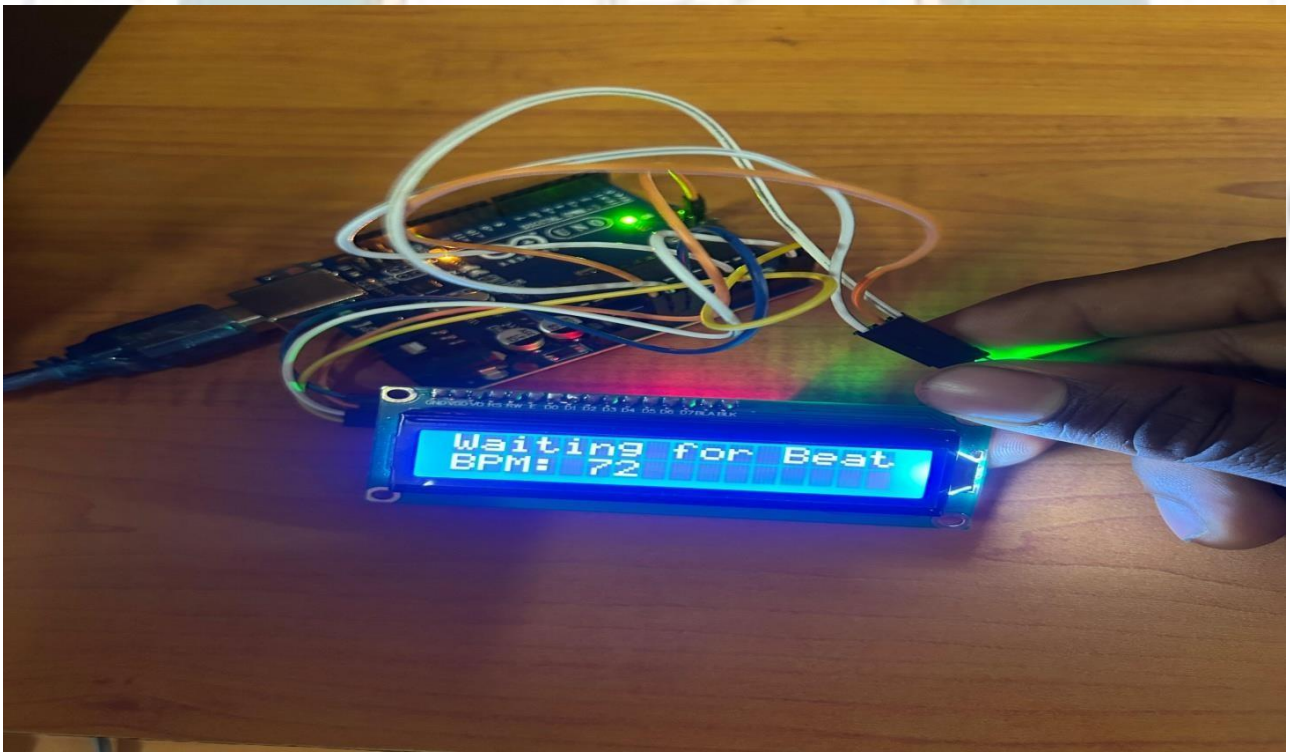
```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
LiquidCrystal_I2C lcd(0x27, 16, 2);

int pulsePin = A0;
int threshold = 550;
int signalValue;

unsigned long lastBeatTime = 0;
unsigned long currentTime;
int BPM = 0;

void setup()
{
  lcd.init();
  lcd.backlight();

  lcd.setCursor(0,0);
  lcd.print("Heartbeat Monitor");

  delay(2000);
  lcd.clear();
}

void loop()
{
  signalValue = analogRead(pulsePin);
  currentTime = millis();

  if(signalValue > threshold)
  {
```

```
if(currentTime - lastBeatTime > 300)
{
  BPM = 60000 / (currentTime - lastBeatTime);
  lastBeatTime = currentTime;

  lcd.clear();
  lcd.setCursor(0,0);
  lcd.print("Heart Rate:");

  lcd.setCursor(0,1);
  lcd.print(BPM);
  lcd.print(" BPM");
}
}
```

DIAGRAM:



OUTPUT:



RESULT:

Thus the above program is verified and executed successfully.